

CLOSE-UP OF PLUTO

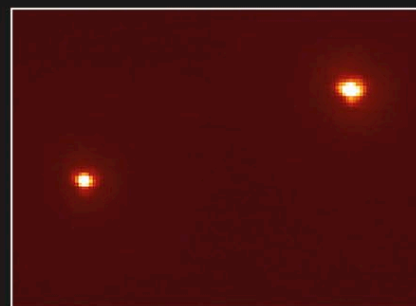
This 200-mile (300-km) wide view looks across Pluto's Tenzing Montes. These are 2–4-mile (3–6-km) high blocks of ice near the edge of the much flatter nitrogen ice that occupies Sputnik Planitia.

**MOONS**

The presence of Pluto's largest moon, Charon, was detected in 1978. Since then, four smaller moons have been discovered orbiting Pluto. Charon is by far the largest of Pluto's moons and is about 750 miles (1,200 km) across. Its mass is about 15 percent that of Pluto. The size and mass of Charon makes it the largest known satellite in relation to its parent body. Charon and Pluto are in a synchronous orbit, meaning they keep the same face toward each other at all times. Both spin on their axes every 6.38 Earth days, and Charon orbits Pluto once in the same period. Because they are so similar in mass, their common center of mass (called their barycenter) about which they each orbit does not lie inside Pluto, but in open space between the two globes.

**PLUTO AND ITS MOONS**

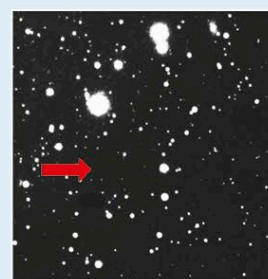
Pluto has five known moons—Charon, Styx, Nix, Kerebros, and Hydra—which are shown here to scale. Charon is by far the largest, with a diameter of about 750 miles (1,200 km). Hydra is about 70 miles (115 km) across, and Nix is about 55 miles (90 km) across. Kerebros and Styx, which were found in 2011 and 2012, are much smaller than the other three moons.

**GROUND-BASED IMAGE**

This image of the Pluto-Charon system was taken by one of the 26.9-ft (8.2-m) telescopes at Paranal Observatory, Chile. Charon was discovered in 1978 by James Christy of the US Naval Observatory, Arizona, who noticed that Pluto's image became elongated periodically. He realized that this was because Pluto has a moon. Charon orbits Pluto at a distance of 10,890 miles (17,530 km).

EXPLORING SPACE**SEARCHING FOR A PLANET**

Pluto was discovered as the result of a deliberate hunt for a "Planet X," which in the early 20th century was thought to affect the orbits of Uranus and Neptune. American astronomer Clyde Tombaugh began his attempt to find this planet at the Lowell Observatory, Arizona, in 1929. His method involved photographing the same area of sky a few days apart and comparing the two images to look for any objects that had moved. On January 23, 1930, Tombaugh took a long exposure of the Delta Geminorum region. On January 29, he imaged the area again, and one "star" in his plates (indicated by the red arrow) had moved. He had discovered Pluto. It later became clear that Pluto was too small to be Planet X, and astronomers today realize there is no need for a Planet X in our models of the solar system.

**PLUTO AND CHARON**

Pluto (below) and Charon (above) are shown here to scale in a composite of New Horizons images. Charon has no atmosphere, and its dramatically fractured surface is dominated by water ice, apart from the red-stained area near the north pole that might be contamination from Pluto. Pluto itself has a fascinatingly varied surface. The bright patch near the middle of the globe, named Sputnik Planitia, is a low-lying region occupied by nitrogen ice that is weak enough to flow and convect even in Pluto's extreme cold. It may have escaped from Pluto's interior to occupy the scar left by a major impact. Elsewhere, the more ancient icy crust is marked by smaller impact craters.